Pavan Kumar Nayakanti

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PROFESSIONAL SUMMARY

Senior SRE and Platform Engineering leader with 22 years driving reliability engineering at FinTech and banking scale. Deep expertise in SLO/SLI/error budget frameworks, chaos engineering (Keptn), fault-tolerant distributed system design, and full-stack observability (Prometheus, Grafana, Dynatrace, Splunk, OpenTelemetry). Reduced MTTR by 40%+ through AIOps-driven automated remediation across ADP, Charles Schwab, BMO, and RBC -- all highly regulated, mission-critical financial platforms. Sustained 99.99% uptime over a 7-year tenure at Microsoft. Experienced presenting SRE strategy and resiliency outcomes directly to VP and C-suite stakeholders. Operates effectively in regulated environments requiring strict compliance with SOX, GDPR, and operational resilience standards.

CORE COMPETENCIES

- SLO/SLI/Error Budget Governance
- Chaos Engineering & Resiliency Testing
- Enterprise Observability Architecture
- Automated Recovery & Toil Reduction
- Fault-Tolerant Distributed System Design
- Disaster Recovery Planning
- DevSecOps & Regulated Compliance
- Kubernetes/OpenShift Platform Operations
- Blameless Postmortem Culture
- Executive-Level SRE Communication

WORK EXPERIENCE

IRIS Software
Sep 2020 - Present

Sr. Specialist - SRE & Platform Engineering Lead | India (Clients: ADP, Charles Schwab, BMO)

- **Architected enterprise observability stack** across ADP's distributed financial platform: Prometheus, Grafana, and Dynatrace providing full-stack metrics, alerting, root-cause analysis, and proactive bottleneck identification -- enabling shift from reactive to predictive operations.
- **Established chaos engineering practice using Keptn** -- designed and executed controlled fault injection scenarios across ADP's production-representative environments, proactively identifying failure modes and hardening platform reliability before incidents occurred.
- **Reduced MTTR by 40%+** through AIOps-driven automated remediation: Ansible + Python workflows triggered by Dynatrace alerts; structured blameless postmortem programme with measurable improvement in change success rate and incident trends.
- **Defined SLO/SLI/error budget frameworks** across multi-client cloud platforms (ADP, Charles Schwab, BMO) -- aligned reliability thresholds to business commitments, enforced error budget policies, and reported SLO attainment to C-suite stakeholders quarterly.
- **Led self-healing and automated recovery architecture**: CI/CD platform (modeas) handling 200+ deployments/hour with automated rollback; Terraform IaC with drift detection and remediation; improving operational resilience to 99.9% across all client platforms.
- Implemented ITIL-aligned incident, problem, and change management across three Fortune 500 banking clients; managed on-call rotations and escalation frameworks for a 35-member SRE team.
- Deployed GenAI-based risk summarisation and Q&A assistants on Azure & AWS for BMO (BCRG/LTS pipelines), reducing manual analysis effort in a regulated financial environment.

IRIS Software
Jun 2018 - Sep 2020

Automation Technology Specialist | Toronto, Canada (Client: Royal Bank of Canada Insurance)

- **Established SRE and DevSecOps principles across 250+ applications** within RBC's DevSecOps programme: observability standards (Splunk-based log aggregation + alerting), SOX/GDPR compliance gates, reliability-focused deployment gates, and automated change evidence pipelines.
- **Designed the Robo Deployment Framework** -- Kubernetes-native CI/CD orchestration with automated fault detection and recovery -- reducing application onboarding from 5 weeks to 5 hours; migrated 250+ applications across 60+ teams to GitHub with zero downtime.
- Governed enterprise-wide change management across 250+ applications -- audit-ready compliance, cross-team incident response, and structured problem management in a regulated insurance environment.

IRIS Software
Feb 2017 - Jun 2018

Automation Architect | Toronto, Canada (Clients: Insurance & Capital Markets)

- Designed fault-tolerant, Kubernetes/Helm-orchestrated deployment pipelines for zero-downtime rollouts in capital markets environments; standardised Python automation frameworks across regulated financial platforms.

Microsoft Corporation
Jul 2007 - Dec 2014

Sr. SDET -- Automation Architect | India

- **Sustained 99.99% platform uptime over 7 years** for Microsoft's global marketing platforms -- proactive SLI monitoring, automated recovery mechanisms, postmortem-driven continuous improvement, and self-healing cloud architectures supporting 200% user growth with zero critical incidents.

TechMahindra -- Technology Manager (2015-2017): Enterprise DevOps transformation, CI/CD adoption, global engineering teams.
Semantic Space / DRDO / NFC (2002-2007): L3 production support, mission-critical systems, software engineering in regulated environments.

KEY PROJECTS

AIOps-Driven Automated Remediation (ADP) `One Touch Recovery`

Designed automated remediation workflows triggered by Dynatrace alerts -- Ansible + Python runbooks covering the most common failure patterns across ADP's financial platform. Reduced MTTR by 40%+; eliminated manual on-call intervention for known failure classes. Paired with blameless postmortem programme to continuously expand runbook coverage.

Dynatrace · Ansible · Python · Prometheus · Grafana · ITIL · Kubernetes

Chaos Engineering Practice (Keptn -- ADP) `Resiliency Testing`

Established production-representative chaos engineering using Keptn: designed fault injection scenarios, automated resiliency validation gates in CI/CD, and tracked reliability trends against SLO thresholds. Proactively identified and remediated failure modes before production impact -- shifting ADP from reactive incident response to predictive reliability operations.

Keptn · Kubernetes · Prometheus · Grafana · Dynatrace · Python · Ansible

Robo Deployment Framework (RBC) `Platform Engineering`

Kubernetes-native CI/CD orchestration platform with automated fault detection, rollback, and compliance gates across 250+ regulated insurance applications. Reduced onboarding from 5 weeks to 5 hours (98% reduction); embedded SOX/GDPR compliance and audit evidence generation into every pipeline.

Kubernetes · Docker · Python · Ansible · Splunk · DevSecOps · SOX/GDPR

EDUCATION

Masters in Artificial Intelligence — Woolf University 2024

General Management Program (GMP) — ISB Hyderabad 2022

M.Sc in Computer Science — Osmania University 2002

CERTIFICATIONS

CKAD (Certified Kubernetes Application Developer, 2025) | **iSRE** Certified SRE | **AZ-305** Azure Solutions Architect Expert (2026) | **AWS SAA-C01** Solutions Architect Associate (2026) | **AI-102** Azure AI Solution Design (2025) | **ITIL** Certificate in IT Service Operations | **PMI-ACP**

TECHNICAL SKILLS

Observability: Prometheus, Grafana, Loki, Mimir, Tempo, Dynatrace, AppDynamics, Splunk, OpenTelemetry, MS AppInsights

Reliability & Resiliency: SLI/SLO, Error Budgets, Chaos Engineering (Keptn), Self-Healing, DR Planning, ITIL, ServiceNow, Blameless Postmortems

Containers & Cloud: Kubernetes, OpenShift, Docker, Helm · AWS, Azure, GCP · Terraform, CloudFormation, Ansible

CI/CD & Security: GitHub Actions, GitLab CI, Jenkins, Azure DevOps, Argo CD · DevSecOps, SOX/GDPR, policy-as-code

Programming: Python, Go, Bash, PowerShell, Groovy · GenAI (Azure & AWS), AIOps, MLOps